

Agricultural Cargo Transport Drone – User Manual

Prepared in IEEE Standard Format

1. System Overview

This drone system is designed for lightweight agricultural logistics and payload delivery. It features a 3D-printed hook-type gripper actuated by a servo motor, controlled via Pixracer flight controller and Radiomaster TX12 ELRS transmitter. The system combines 2300 KV BLDC motors with 5-inch tri-blade propellers, optimized for efficient lift during small-load transportation.

2. Components List

Component	Specification	Function
Frame	Skyris Quadcopter	Main structural base
Flight Controller	Pixracer	Central flight management
Transmitter	Radiomaster TX12 ELRS	Pilot control
Receiver	ELRS 2.4 GHz	Communication link
Motors	2300 KV BLDC	Propulsion system
Propellers	5-inch Tri-Blade	Thrust generation
Servo	5V Metal Gear	Gripper actuation
Gripper	3D Printed Hook	Payload pickup and release
Battery	4S 2300 mAh Li-Po	Power supply

The following table lists all the components used in the Agricultural Cargo Drone system.

3. Electrical Connections

Servo: Connected to Pixracer AUX1 (Channel 8). Power is supplied via 5V BEC. The Radiomaster TX12 ELRS controls the servo via Channel 8. Motors connect to ESCs, and ESCs connect to Pixracer outputs 1–4. Power is provided by a 4S 2300 mAh Li-Po battery.

4. Firmware Configuration (Pixracer)

The Pixracer is configured via Mission Planner or QGroundControl. Frame type is set to X. Servo8 is assigned RC pass-through function, with PWM range 1000–2000 μ s. Radio calibration and ESC calibration are performed before flight.

5. Mechanical Assembly

The servo and gripper assembly are mounted at the center-bottom of the Skyris frame. Motors are fixed at each arm end with tri-blade propellers. The battery and Pixracer are placed at the center to maintain CG balance.

6. Pre-Flight Checklist

Ensure propellers are secure, servo motion verified, battery charged, gripper attached, and transmitter linked. Verify failsafe and stabilization mode before arming.

7. Operation Procedure

Power on transmitter and drone, test servo actuation, attach payload, take off, navigate to delivery point, release payload using Channel 8 switch, and land safely.

8. Safety Guidelines

Keep clear from propellers, avoid payloads over 500 g, fly in open areas, perform calibration regularly, and use proper failsafe settings.

9. Maintenance

Inspect motors, servo, and propellers regularly. Keep firmware updated. Store Li-Po batteries at 3.8V per cell. Replace damaged parts immediately.

10. Troubleshooting

Refer to the troubleshooting table for resolving common operational issues.

11. Future Enhancements

Incorporate load sensors, autonomous drop functionality, FPV camera, and improved brushless gripper actuator.

Issue	Cause	Solution
Servo not responding	Incorrect channel mapping	Reassign switch and check 5V connection
Servo jittering	Weak power or noise	Use separate BEC or capacitor
Payload not releasing	Low servo travel	Adjust PWM endpoints
Unstable flight	Unbalanced payload	Recenter load and retune PIDs

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